

Whole-Number Division: Interpreting Remainders

Deciding what the remainder means: drop it, round up, or report what is left over

- **Drop it** — the question asks how many *complete* groups you can make. Leftovers do not make a whole group, so the quotient alone is the answer.
- **Round up** — the question asks how many groups you *need* so that nothing is left behind. The leftovers still need one more group, so add 1 to the quotient.
- **Report the leftover** — the question asks how many are left over, or how many are in the last partly filled group. The remainder itself is the answer.

Directions: For every problem, write the division you used (total $\div$ group size), write the quotient and the remainder, then decide which of the three meanings above the question is asking for. Answer with the number *and* what it counts.

1. A farm stand has **53** apples. Each bag holds **6** apples. The stand only sells *full* bags. How many full bags can be made?

Answer: _____

2. Devon has **47** stickers and shares them equally among **5** friends, giving each friend as many whole stickers as possible. How many stickers are left over?

Answer: _____

3. A school is taking **128** students on a trip. Each van seats **9** students. Every student must have a seat. How many vans are needed?

Answer: _____

4. A ribbon is **275** centimetres long. It is cut into pieces that are each **8** centimetres long. How many whole pieces can be cut?

Answer: _____

5. A bakery has **350** eggs. Each carton holds **6** eggs. How many cartons are needed so that every egg is in a carton?

Answer: _____

6. Rosa is putting **623** trading cards into an album. Each page holds **9** cards. She fills each page before starting the next one. How many cards are on the last page she uses?

Answer: _____

7. A field trip takes **237** students. Each bus seats **40** students. Buses are filled one at a time before the next one is used.

- (a) **Estimate first.** Round the number of students to the nearest ten, then divide by the number of seats. About how many buses will be needed?

Answer: _____

- (b) How many students ride on the last bus?

Answer: _____

8. A warehouse packs **1248** pencils into boxes. Each box holds **9** pencils. How many boxes can be completely filled?

Answer: _____

9. A library is putting **2455** books onto shelves. Each shelf holds **8** books, and shelves are filled one at a time.

(a) How many shelves are needed so that every book is on a shelf?

Answer: _____

(b) How many books are on the last shelf?

Answer: _____

10. Find and fix the mistake. Maya has **314** marbles and boxes that each hold **6** marbles. She divided correctly and wrote $314 \div 6 = 52 \text{ R } 2$, then answered: “**52** boxes are needed to hold all the marbles.”

- (a) Maya’s division is right but her answer is wrong. How many boxes are really needed so that every marble is in a box?

Answer: _____

- (b) How many marbles are in the last box?

Answer: _____